

Machine Learning Approaches to Predict Poor Mental Health Among U.S. Adults: Evidence from BRFSS 2024

1. Some suggested Variables for Machine Learning Models

Category	Variable Name	Description	Type
Target Variable	MENTHLTH	Number of poor mental health days (0–30)	Continuous → Binary
	mental_risk	≥14 poor mental health days (0 = No, 1 = Yes)	Binary
Behavioral	EXERANY2	Participation in physical activity	Binary
	_TOTINDA	Meets physical activity guidelines	Binary
	_SMOKER3	Smoking status	Categorical
	_DRNKWK3	Number of alcoholic drinks per week	Continuous
Physiological	_BMI5	Body Mass Index (BMI × 100)	Continuous
	_BMI5CAT	BMI category (Underweight–Obese)	Categorical
	PHYSHLTH	Number of poor physical health days	Continuous
Lifestyle	SLEPTIM1	Average hours of sleep	Continuous
Demographic	_AGE80	Age	Continuous
	SEXVAR	Sex	Binary
	EDUCA	Education level	Categorical

	INCOME2	Income category	Categorical
	EMPLOY1	Employment status	Categorical
Healthcare	HLTHPLN1	Health insurance coverage	Binary
	PERSDOC2	Personal doctor availability	Categorical
	CHECKUP1	Time since last medical checkup	Categorical

2. List of Machine Learning Models

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|---------------------------------------|----------------|
| 1. Logistic Regression (base line) | Classification |
| 2. Decision Tree | Classification |
| 3. Random Forest (Highly recommended) | |
| 4. XGBoost | |

3. Project Guidelines

1. Objective
 - Predict mental health risk
2. Download BRFSS 2024 dataset from CDC website
3. Read data file in Python or R
4. Select required variables
5. Data Cleaning
 - Handle missing values
 - Normalize variables
6. Exploratory Data Analysis (EDA)
 - Examine Correlations between variables
 - Distributions
7. Modeling
 - Train/test split of cleaned data
 - Run multiple models (model listed above)
8. Evaluation of Model Performance using the following measures
 - Accuracy
 - ROC-AUC
 - Confusion matrix
9. Interpretation
 - Feature importance ---- which features (or predictors) are important in predicting mental health risk

4. Write a paper